

Joo Wang Lee

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EDUCATION

Oregon State University

September 2022 – June 2026

Bachelor of Science, Majoring in Applied Computer Science: Data Science

- Honor Roll | Dean's List
- Relevant Coursework: Data Structures | Analysis of Algorithms | Computer Architecture | Operating Systems | Database Management Systems | Software Engineering | Web Development | Cybersecurity

SKILLS

Programming: C, C++, Python, Raku, HTML, JavaScript, CSS, SQL, Wireshark, R, AWS, Terraform

Database: MySQL, Oracle SQL, Elasticsearch, DynamoDB, Redis, Kafka, Tableau

Language: Bilingual (Native in Korean & English)

PROJECTS

API Library – Full-Stack Flask App with Cloud Deployment

July 2026 – August 2026

- Built a RESTful API (Flask + SQLAlchemy) with a normalized Postgres schema for books, categories, customers, and loans, backing both customer-facing UI and an internal admin panel
- Implemented session-based authentication with hashed passwords (Werkzeug), including a separate self-registration flow distinct from staff-managed customer records
- Wrote a pytest test suite (5 modules covering auth, books, customers, loans) and containerized the app with Docker/Docker Compose for local and production parity
- Provisioned production AWS infrastructure from scratch with Terraform – EC2, RDS Postgres, IAM roles/instance profiles, and Secrets Manager – using an S3 + DynamoDB remote backend for shared, locked state
- Built a CI/CD pipeline with GitHub Actions that auto-deploys to EC2 via SSH on push to main, gated behind a manual approval environment

3D Elastic Properties of Biomaterials from Molecular Dynamics Simulations (https://github.com/vanegasj/mdstress-library/tree/capstone_optimization)

September 2025 – June 2026

- Optimized and parallelized molecular dynamics post-processing code (MDStress) using profiling, vectorization (SIMD), GPU acceleration, and multi-node MPI to reduce runtime and improve scalability
- Designed automated unit and regression testing framework to ensure numerical consistency across code versions
- Implemented and validated higher-order interatomic potential derivatives using symbolic algebra and automatic differentiation for elasticity and stress-fluctuation analysis
- Ported and modernized GROMACS-LS modifications to newer GROMACS releases to leverage improved performance features and sampling efficiency

Transaction Fraud Detector

August 2025

- Built an end-to-end ML fraud detection pipeline in Python, achieving an F1 score of 0.993 and ROC-AUC of 0.9998 on a held-out test set across 50,000 synthetic transactions
- Engineered 23 features from raw transaction data including log-transformed amounts, location deviation, and velocity signals, and handled severe class imbalance (97/3%) using SMOTE oversampling
- Trained and benchmarked four classifiers (Logistic Regression, Decision Tree, Random Forest, Gradient Boosting), selecting the best model using recall-optimised evaluation to minimise missed fraud
- Designed a hybrid decision engine combining ML probability scoring with a 9-rule deterministic layer for instant blocking of high-confidence fraud cases
- Deployed predictions via a Flask REST API with real-time single-transaction and batch CSV scoring endpoints

Database Cleaning and Analysis

March 2024

- Integrated data from multiple sources, including SQL databases and Python pipelines
- Cleaned and preprocessed datasets in Python and R, addressing missing values, outliers, and structural inconsistencies

- Conducted data manipulation and feature engineering to prepare datasets for statistical analysis and modeling
- Produced clear visualizations and dashboards with matplotlib, seaborn, and related tools to summarize insights and trends

Small Shell

January 2024

- Built a custom Linux-style shell in C supporting command execution, background processes, signal handling, I/O redirection, and a foreground-only mode

EXPERIENCE

Holiday Inn Express

Front Desk Receptionist | May 2026 - Present

- Managed guest check-in and check-out processes efficiently, ensuring a smooth and welcoming experience
- Maintained and updated reservation systems, tracking bookings, cancellations, and room availability
- Accurately recorded and updated guest information, ensuring data integrity and confidentiality
- Demonstrated strong multitasking and organizational skills in a fast-paced environment
- Processed payments, verified billing information, and maintained accurate financial records
- Assisted in managing overbookings and scheduling adjustments to optimize occupancy rates

RiseStack – Internship

July 2026 - Present

- Built the data-loss-prevention engine for an AI gateway that proxies enterprise LLM traffic
- Implemented four enforcement modes (block, flag, redact, and reversible tokenization)
- Owned end-to-end QA for the platform to test 50+ edge cases utilizing Postman

ACTIVITIES

Teaching Assistant,

St Andrew Kim's Korean Catholic Church | Seattle, WA

- Assisted in caring for children with disabilities every Sunday
- Foster an inclusive environment for kids to learn and play together

OSU App Club

- Collaborate with peers to design and develop mobile applications
- Engage in code reviews and discussions to improve software quality and maintainability
- Contribute to group app projects by implementing features, debugging issues, and refining app performance